1.
Connect a USB webcam to your computer, install any required drivers, and verify that the webcam works by opening the default camera app and capturing a photo.

. Connecting a USB Webcam and Testing It on a Computer

Aim: To connect a USB webcam to a computer, install the required drivers, and verify its working by opening the Camera app and capturing a photo.

Step-by-Step Procedure

Step 1: Connect the USB Webcam

- Take a USB webcam and connect its USB cable to an available USB port on the computer.
- Place the webcam securely on top of the monitor or another stable surface.
- Ensure that the USB connector is properly inserted.

Step 2: Install the Required Drivers

- Windows may automatically detect the webcam and install the required drivers.
- If the webcam does not work, open **Device Manager** and check under **Cameras** or **Imaging devices**.
- If a driver is missing, download the appropriate driver from the webcam manufacturer's official website and install it.

Step 3: Allow Camera Access

- Open **Settings → Privacy → Camera** in Windows 10.
- Turn on camera access for the device.
- Allow desktop applications to access the camera if required.

Step 4: Open the Camera App

- Click the **Start** menu.
- Search for **Camera**.
- Open the Camera app.
- If prompted, allow the application to access the webcam.

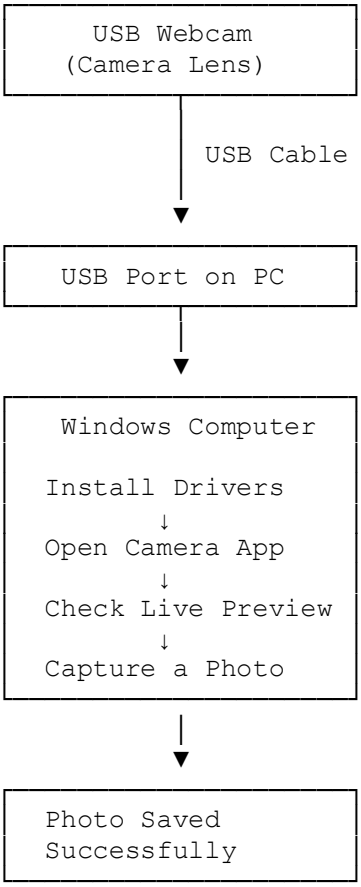
Step 5: Verify the Webcam

- Check whether a live video preview appears on the screen.
- Adjust the webcam's position and check that the image is clear.
- If the screen remains black, check the USB connection, camera permissions, and driver status.

Step 6: Capture a Photo

- Position yourself or an object in front of the webcam.
- Click the **Take Photo** button in the Camera app.
- Open the saved photo to confirm that the webcam has captured the image successfully.

Diagram: USB Webcam Connection



Expected Result

S.No.	Action	Expected Result
1	Connect the webcam	The computer detects the webcam.
2	Install the driver	The webcam becomes ready to use.
3	Open the Camera app	A live video preview appears.
4	Capture a photo	The photo is saved successfully.

2.

Set up a local printer (USB or network), install the correct driver, and print a test page to confirm successful installation.

Setting Up a Local Printer and Printing a Test Page

Aim: To connect a printer to a computer, install the correct printer driver, and print a test page to verify that the printer is working properly.

Requirements:

- Computer or laptop
- Printer
- USB cable or Wi-Fi network
- Printer driver
- Paper and ink or toner

Step-by-Step Procedure

Step 1: Connect the Printer

- Place the printer on a flat surface near the computer.
- Connect the printer to a power socket and switch it on.
- **USB connection:** Connect the USB cable to the printer and computer.
- **Network connection:** Connect the printer and computer to the same Wi-Fi network, or connect the printer to the local network using an Ethernet cable if supported.

Step 2: Install the Correct Printer Driver

- Visit the printer manufacturer's official website.
- Search for the printer's exact model.
- Download the compatible driver for your operating system.
- Run the downloaded installer and follow the on-screen instructions.

Note: Windows may automatically install the appropriate driver when the printer is connected.

Step 3: Add the Printer

For Windows 10:

1. Open **Settings**.
2. Select **Devices**.
3. Click **Printers & scanners**.
4. Select **Add a printer or scanner**.
5. Wait for Windows to detect the printer.
6. Select the correct printer and click **Add device**.

If the printer does not appear, check its connection and follow the manufacturer's setup instructions.

Step 4: Check the Printer Status

- Open **Settings → Devices → Printers & scanners**.
- Select your printer.
- Click **Manage**.
- Check that the printer is installed and available.
- Ensure that paper is loaded and ink or toner is sufficient.

Step 5: Print a Test Page

1. Open **Control Panel**.
2. Select **Devices and Printers**.
3. Right-click the installed printer.
4. Select **Printer properties**.
5. Click **Print Test Page**.
6. Wait for the printer to print the page.

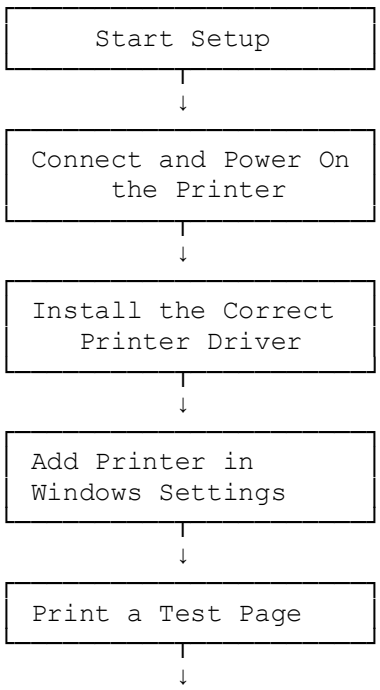
Step 6: Verify the Result

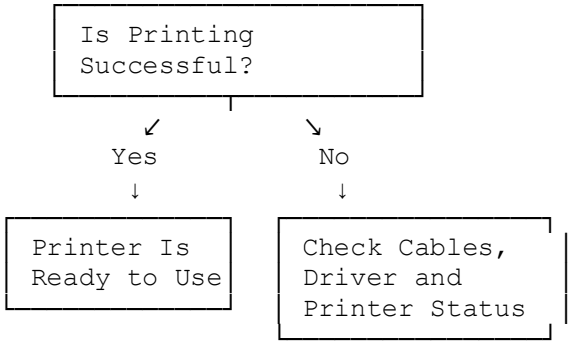
Check the printed test page for:

- Clear and readable text.
- Correct alignment.
- Proper colors, if applicable.
- No missing lines or unexpected blank areas.

If the test page prints successfully, the printer has been installed and is ready to use.

Diagram: Printer Setup Process





Expected Result

S.No.	Task	Expected Result
1	Connect the printer	The printer powers on and connects to the computer or network.
2	Install the driver	The correct printer driver is installed.
3	Add the printer	The printer appears in the printer settings.
4	Print a test page	The printer produces a test page successfully.

3.

Simulate a connectivity issue by unplugging the printer's USB cable, then use Windows Device Manager to identify and resolve the 'printer not detected' problem.

Hint: Check for error icons and use the 'Scan for hardware changes' option after reconnecting.

Troubleshooting a Printer Not Detected Using Windows Device Manager

Aim: To simulate a printer connectivity issue by unplugging its USB cable, identify the problem using Windows Device Manager, and restore the connection.

Requirements:

- Computer or laptop running Windows
- USB printer
- USB cable
- Printer driver

Step-by-Step Procedure

Step 1: Disconnect the Printer

1. Turn on the computer and printer.
2. Unplug the printer's USB cable from the computer.
3. The computer may no longer detect the printer.
4. This simulates a printer connectivity problem.

Step 2: Open Device Manager

1. Right-click the **Start** button.
2. Select **Device Manager**.
3. Device Manager displays the hardware devices connected to the computer.

Step 3: Check for the Printer

1. Expand **Printers** or **Print queues**.
2. Look for the printer's name.
3. Also check **Universal Serial Bus controllers** for any USB-related warning icons.
4. A yellow exclamation mark may indicate a device or driver problem.

Note: When the USB cable is disconnected, the printer may simply disappear from Device Manager. This does not necessarily indicate a driver error.

Step 4: Reconnect the USB Cable

1. Reconnect the USB cable securely to the computer and printer.
2. Ensure that the printer is switched on.
3. Wait a few seconds for Windows to detect the device.

Step 5: Scan for Hardware Changes

1. In Device Manager, click the **Action** menu.
2. Select **Scan for hardware changes**.
3. Windows will search for newly connected hardware.
4. Check whether the printer appears under **Printers**, **Print queues**, or another relevant device category.

Step 6: Check the Driver

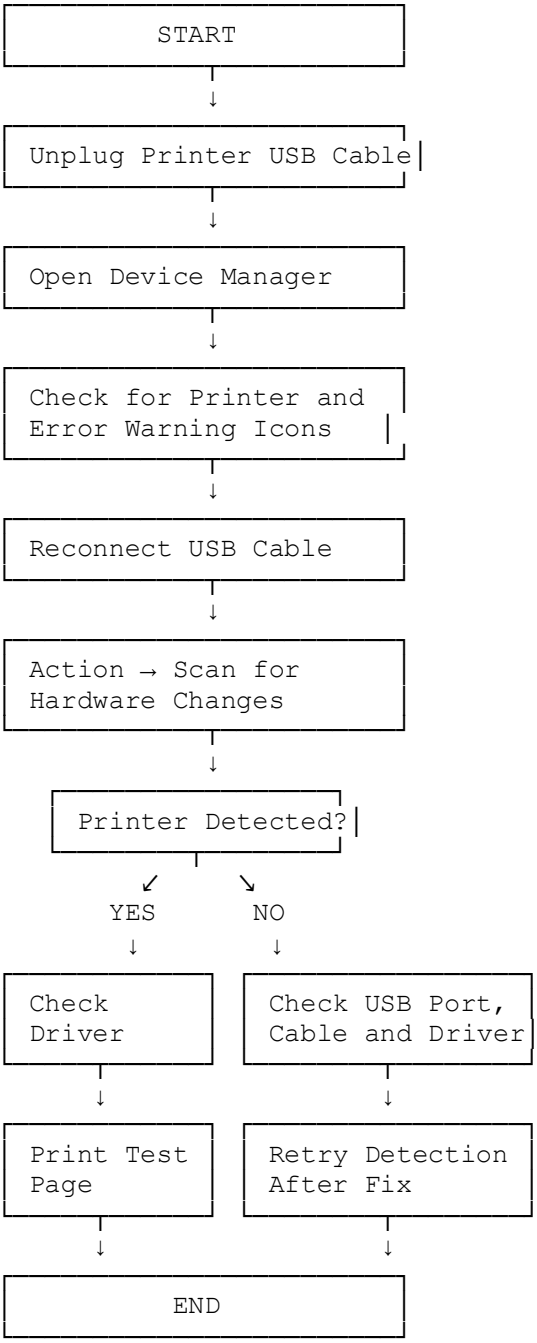
If the printer is still not detected:

1. Right-click the printer or the relevant device with a warning icon.
2. Select **Update driver**.
3. Follow the on-screen instructions.
4. If necessary, download and install the correct driver from the printer manufacturer's official website.

Step 7: Verify the Connection

1. Open **Settings → Devices → Printers & scanners**.
2. Check whether the printer appears in the list.
3. Select the printer and open its management options.
4. Print a test page to confirm that the connection is working.

Diagram: Troubleshooting Flowchart



Expected Results

S.No.	Action	Expected Result
1	Unplug the USB cable	The computer may stop detecting the printer.
2	Open Device Manager	Connected hardware and possible warning icons are visible.
3	Reconnect the USB cable	Windows may automatically detect the printer.
4	Scan for hardware changes	Windows searches for connected hardware again.
5	Check the driver	Driver problems can be identified and addressed if present.

S.No.	Action	Expected Result
6	Print a test page	Successful printing confirms that the connection has been restored.

4.

A user reports that their webcam video is not displaying in Zoom, but it works in the default camera app. Troubleshoot and resolve the issue so the webcam works in both applications.

Hint: Check if another app is using the webcam or if Zoom permissions are enabled.

Troubleshooting a Webcam That Works in the Camera App but Not in Zoom

Aim: To identify and resolve the problem when a webcam works in the Windows Camera app but does not display video in Zoom.

Problem: The webcam is functioning correctly, but Zoom cannot display its video.

Requirements:

- Computer or laptop running Windows
- Webcam
- Zoom application
- Internet connection

Step-by-Step Troubleshooting Procedure

Step 1: Check the Webcam in the Camera App

1. Open the Windows **Camera** app.
2. Check whether the webcam displays a live video preview.
3. If the video works, close the Camera app completely.

Reason: This confirms that the webcam is functioning and helps identify whether the problem is specific to Zoom.

Step 2: Close Other Applications Using the Webcam

1. Close the Camera app and other video-calling applications, such as Microsoft Teams or Skype.
2. Check whether any other application is using the webcam.
3. Reopen Zoom and test the video.

Reason: Some applications may prevent other applications from accessing the webcam at the same time.

Step 3: Check Windows Camera Privacy Settings

1. Open **Settings** in Windows.
2. Select **Privacy → Camera**.
3. Ensure that camera access for the device is turned on.
4. Enable **Allow apps to access your camera**, if available.
5. Enable **Allow desktop apps to access your camera**.

Reason: Zoom needs permission to access the webcam.

Step 4: Select the Correct Camera in Zoom

1. Open Zoom.
2. Click your profile picture and select **Settings**.
3. Open the **Video** or **Video & effects** section, depending on the Zoom version.
4. Find the **Camera** dropdown menu.
5. Select the correct webcam instead of another camera or virtual camera.
6. Check whether the video preview appears.

Reason: Zoom may be using the wrong camera device.

Step 5: Check Zoom's Camera Permissions and Settings

1. Ensure that the camera is enabled in Zoom.
2. If you are already in a meeting, click **Start Video**.
3. Check whether the video preview appears in Zoom's settings.
4. If your computer is managed by an organization, check whether any administrator restrictions prevent camera access.

Reason: The camera may be available in Windows but disabled or restricted in Zoom.

Step 6: Restart Zoom and the Computer

1. Close Zoom completely.
2. Reopen Zoom and check the video preview.
3. If the issue continues, restart the computer.
4. Open Zoom again and test the webcam.

Reason: Restarting can resolve temporary application or device-access problems.

Step 7: Update Zoom and the Webcam Driver

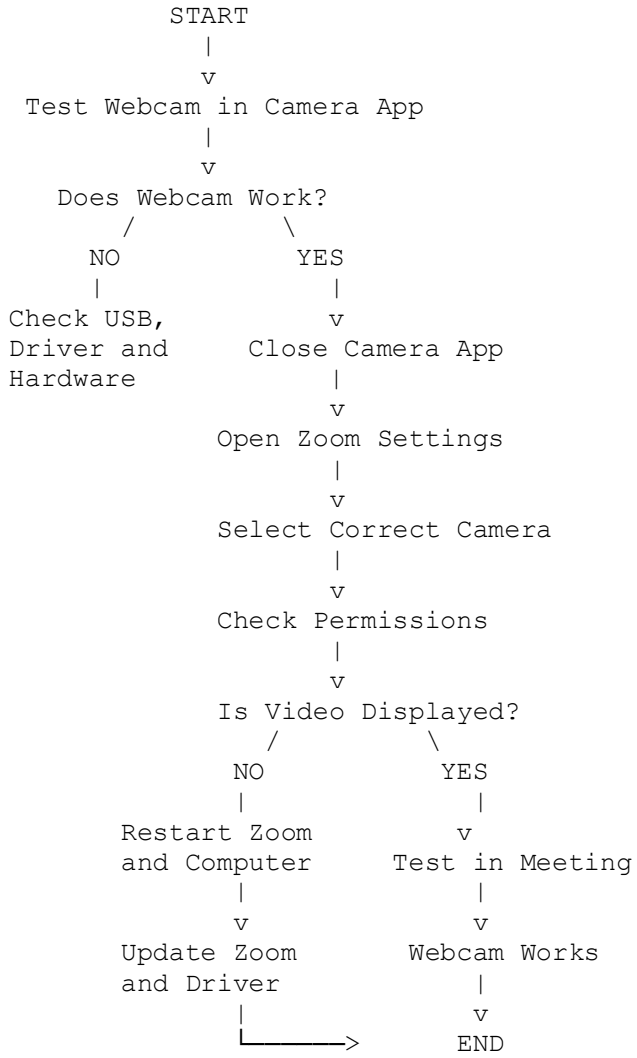
1. Check for Zoom updates using its profile menu.
2. Open **Device Manager** in Windows.
3. Expand **Cameras** or **Imaging devices**.
4. Right-click the webcam and select **Update driver** if necessary.
5. Restart Zoom and test the video again.

Reason: Outdated software or driver issues can interfere with webcam functionality.

Step 8: Verify the Final Result

1. Open the Windows Camera app and confirm that the webcam works.
2. Close the Camera app.
3. Open Zoom and navigate to **Settings → Video**.
4. Confirm that the correct camera is selected and the video preview appears.
5. Join a test meeting and check that other participants can see the video.

Troubleshooting Flowchart



Expected Results

S.No.	Troubleshooting Action	Expected Result
1	Test the webcam in the Camera app	Confirms that the webcam is working.
2	Close other camera applications	Releases the webcam for Zoom to use.
3	Enable Windows camera permissions	Allows desktop applications to access the webcam.
4	Select the correct camera in Zoom	Displays video from the intended webcam.
5	Restart Zoom or the computer	Resolves temporary application issues.

S.No.	Troubleshooting Action	Expected Result
6	Update software or drivers	Addresses relevant compatibility or driver problems.